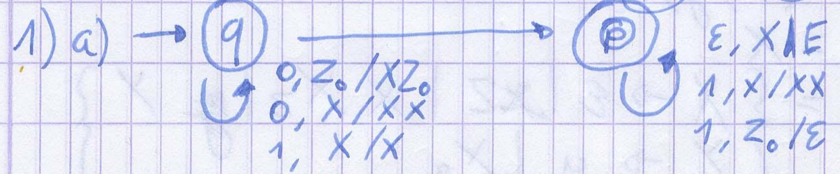
Machines & Berekenbaarheid - Oefeningenreeks 4

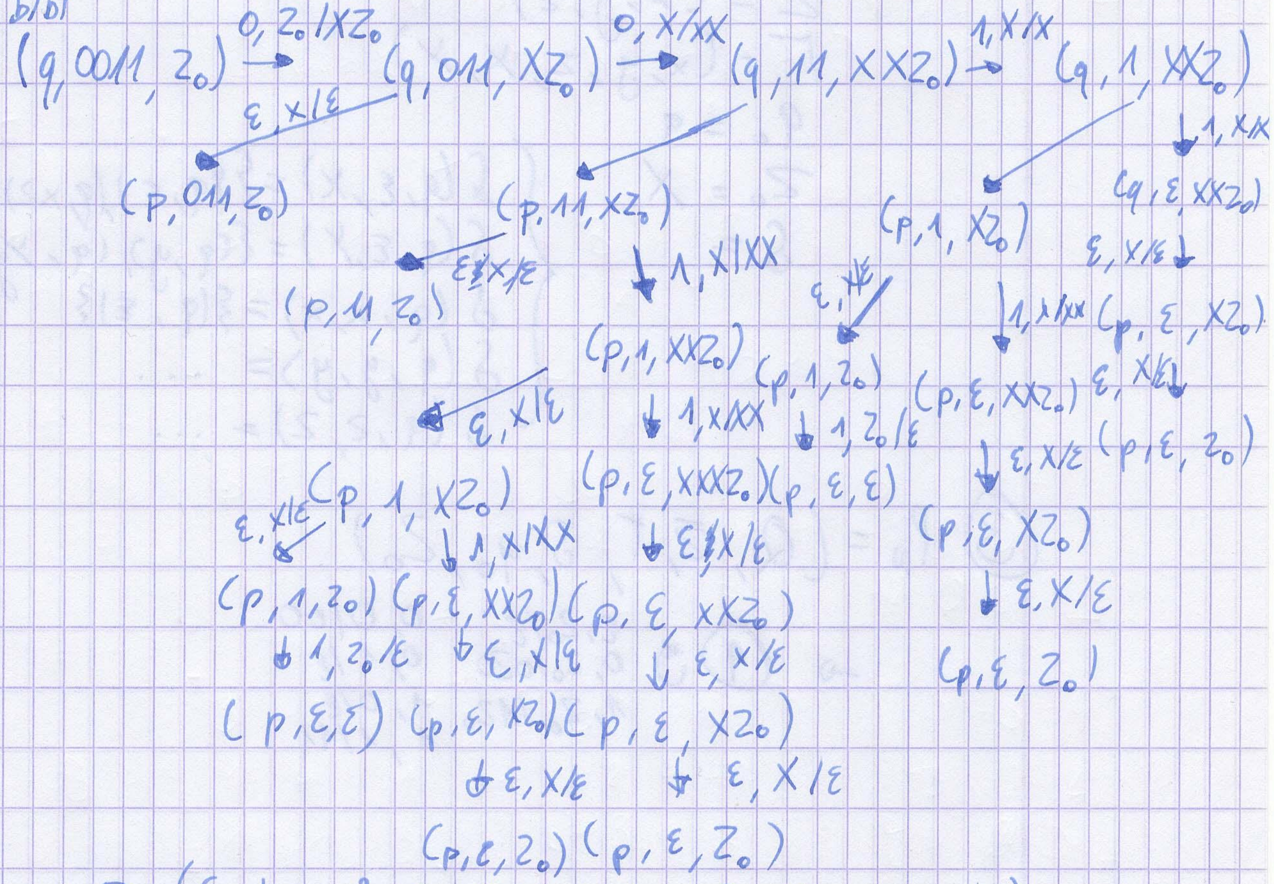
1) a)



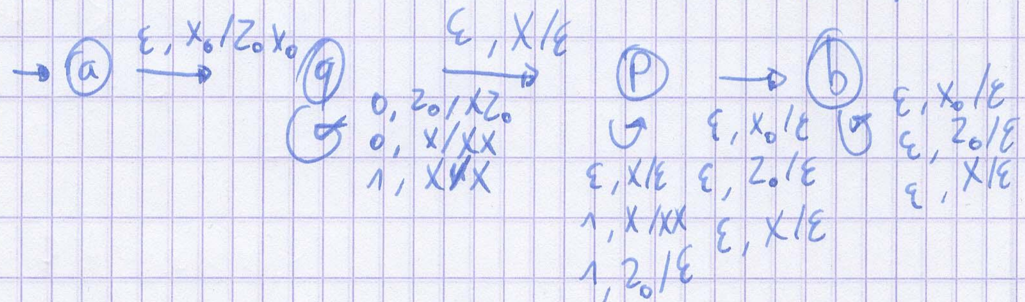
b) a) $(q, 01, Z_0) \xrightarrow{0, Z_0/XZ_0} (q, 1, XZ_0) \xrightarrow{\varepsilon, X/\varepsilon} (p, 1, Z_0) \xrightarrow{1, Z_0/\varepsilon} (p, \varepsilon, \varepsilon)$

$(q, 1, XZ_0) \xrightarrow{1, X/X} (q, \varepsilon, XZ_0) \xrightarrow{\varepsilon, X/\varepsilon} (p, \varepsilon, Z_0)$

b) b)



c) $P_N = (\{a, b, q, p\}, \{0, 1\}, \{X, X_0, Z_0\}, \delta_N, a, X_0)$



CFG

(2) $G = (V, T, P, S)$

$T = \{x, y, z\}$

$S = X$

$P = \left\{ \begin{array}{l} X \to \varepsilon \mid xz \mid xXz \mid \cancel{y}\ Y \\ Y \to y \mid Xy \end{array} \right\}$

$V = \{X, Y\}$

$P = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$

$Q = \{q\}$

$\Sigma = (x, y, z)$

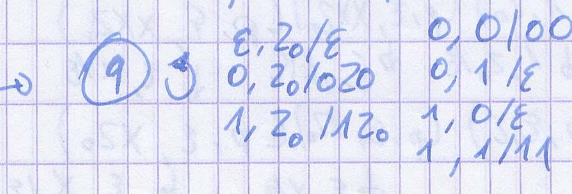
$\Gamma = (x, y, z, X, Y)$

$q_0 = q$

$Z_0 = X$

$\delta = \left\{ \begin{array}{l} \delta(q, \varepsilon, X) = \{(q, \varepsilon), (q, xz), (q, xXz), (q, Y)\} \\ \delta(q, \varepsilon, Y) = \{(q, y), (q, Xy)\} \\ \delta(q, x, x) = \{(q, \varepsilon)\} \\ \delta(q, y, y) = \ldots \\ \delta(q, z, z) = \ldots \end{array} \right.$

(3) $P_N = (Q, \Sigma, \Gamma, \delta, q_0, Z_0)$



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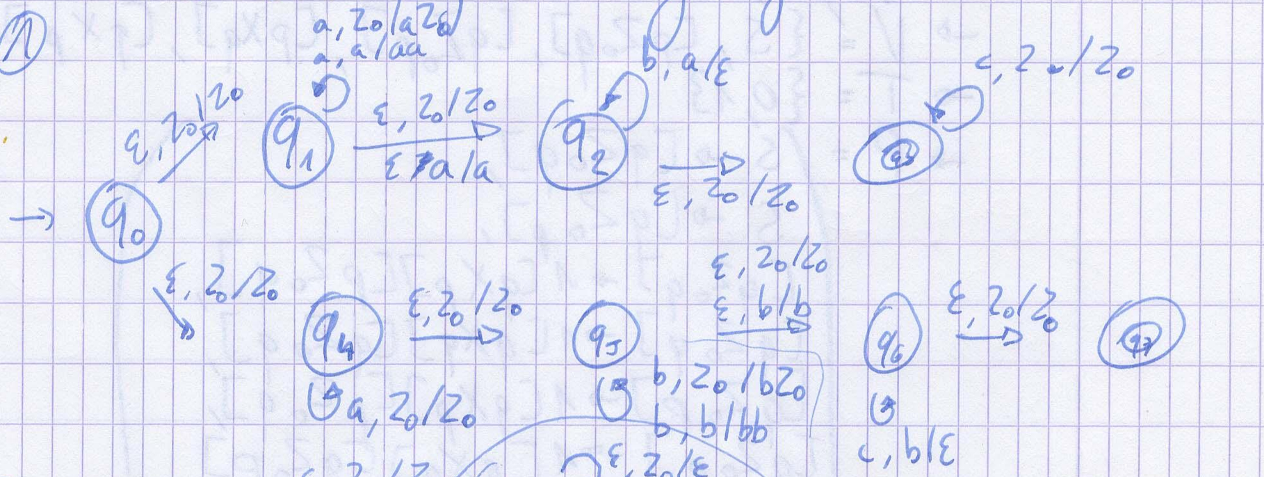
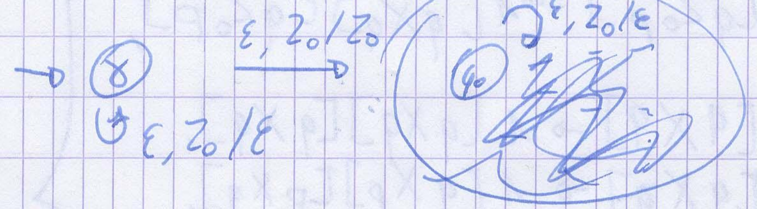
(1)

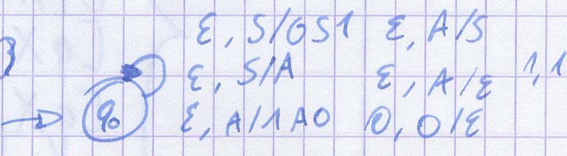


(2)



(3) $\forall A \in V: \delta(q, \varepsilon, A) = \{(q, \beta) : A \to \beta \in \rho\}$

$\forall a \in T: \delta(q, a, a) = (q, \varepsilon)$



(4) $P_N = (Q, \Sigma, \Gamma, \delta, q_0, Z_0) \to G = (V, \Sigma, R, S)$

$\to V = \{[pXq] : \{p, q\} \subseteq Q, X \in \Gamma \cup \{S\}$

$\to R = \{S \to [q_0 Z_0 p] : p \in Q\}$

$[qXv_k] \to a[vY_1v_1] \dots [v_{k-1}Y_kv_k] : a \in \Sigma \cup \{\epsilon\}$, $\bigcup_{n=1}^{k} v_n \subseteq Q$, $(v, \bigcup_{n=1}^{k} Y_n) \in \delta(q, a, X)$

$[qXv] \to a : a \in \Sigma \cup \{\epsilon\}, (v, \epsilon) \in \delta(q, a, X)$

ZOZ

$G = (V, T, P, S)$

$\rightarrow V = \{S, [pZ_0q], [qZ_0p], [pXq], [qXp]\}$

$\rightarrow T = \{0, 1\}$

$$
\rightarrow P = \left\{
\begin{array}{l}
S \rightarrow [qZ_0q], \\
S \rightarrow [qZ_0p], \\
[qZ_0q] \rightarrow 1[qXp][pZ_0q], \\
[qZ_0q] \rightarrow 1[qXq][qZ_0q], \\
[qZ_0p] \rightarrow 1[qXp][pZ_0q], \\
[qZ_0p] \rightarrow 1[qXq][qZ_0p], \\
\\
[qXq] \rightarrow 1[qXq][qXq], \\
[qXq] \rightarrow 1[qXp][pXq], \\
[qXp] \rightarrow 1[qXp][qXp], \\
[qXp] \rightarrow 1[qXp][pXp], \\
\\
[qXq] \rightarrow \varepsilon \\
[pXp] \rightarrow \varepsilon \\
\\
[pZ_0p] \rightarrow 0[qZ_0p], \\
[pZ_0q] \rightarrow 0[qZ_0q]
\end{array}
\right\}
$$

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①

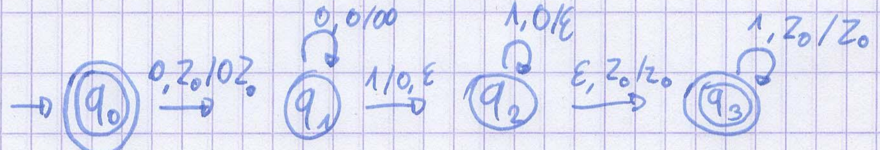
a) A PDA $P = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$ deterministisch

$\Longleftrightarrow$

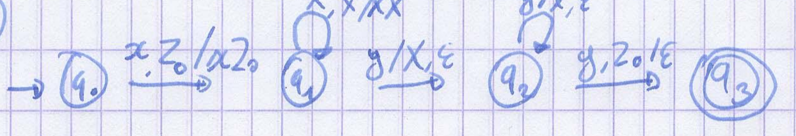
1) $|\delta(q, a, X)| \leq 1$

2) Als $|\delta(q, a, X)| \neq 0$ dan $\delta(q, \epsilon, X) = \emptyset$

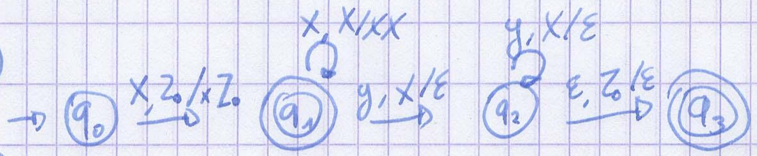
b)



②



③ a)



b)

Nee, PDA's die de 0-stack accepteren, herkennen slechts talen met "prefix"

Deze taal heeft deze niet